

Applied Mathematics I

CHAPTER 6: Integrals

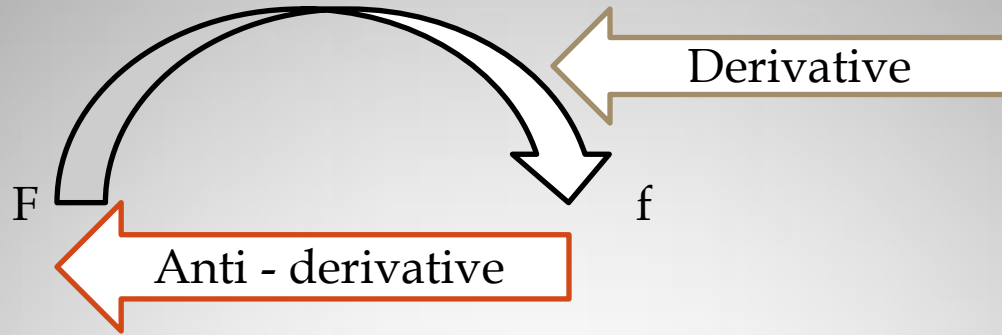
5.1 Anti derivative

The derivation of a given function always gives another function

Given a function $F(x)$, we can find another function $f(x)$ such that

$$F'(x) = f(x)$$

then f is called the derivative of $F(x)$ and $F(x)$ is called anti-derivative of $f(x)$



But also

$(F + c)' = F' + C' = F' = f$ this implies $F + c$, where c is any arbitrary constant, is the general anti derivative of f

Example: Find the anti derivative of the function.

a. $f(x) = 2$ b. $f(x) = x$ c. $f(x) = x + 3$

d. $f(x) = x^2 + 2x$ e. $f(x) = x^3 + 3x^2 + x$

Solution.

- a. We must find function whose derivative is $f(x) = 2$
2x is the function whose derivative is 2 and the only other function whose derivative is 2 are $2x + c$
then we take $2x + c$ as the anti - derivative of $f(x) = 2$, where c is any constant
- b. We must find here also function whose derivative is $f(x) = x$ and $\frac{1}{2}x^2$ is the function whose derivative is x and the only other function whose derivative is x are $\frac{1}{2}x^2 + c$ then we take $\frac{1}{2}x^2 + c$ as the anti -derivative of $f(x) = x$, where c is any constant

c. In here we must find function whose derivative is $f(x)=x+3$
 $\frac{1}{2}x^2 + 3x$ is a function whose derivative is $x + 3$ and the
the only other functions whose derivative is $x+3$ are
 $\frac{1}{2}x^2 + 3x + c$ so that we take $\frac{1}{2}x^2 + 3x + c$ as the
antiderivative of $f(x) = x+ 3$, where c is a constant.

d and e are left as exercise

5.2 Partition, Lower sum, Upper sum and Riemann sum

Definition: A partition of $[a, b]$ is a finite set P of points
 $x_0, x_1, x_2, \dots, x_n$ such that $a = x_0 < x_1 < x_2 < \dots < x_n = b$ and
we describe p as $p = \{x_0, x_1, x_2, \dots, x_n\}$

Note that any partition of $[a, b]$ must contain both end points a
and b

Example: The partition of $[0,2]$ can be either of the following.

$$\left\{0, \frac{1}{2}, 1, \frac{3}{2}, 2\right\} \quad ; \{0,2\}; \left\{0, \frac{1}{2}, \frac{\pi}{4}, 1, \frac{\pi}{2}, \frac{3\pi}{5}, 2\right\} \quad or \quad \left\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2\right\}$$

But not $\left\{\frac{1}{2}, 1, \frac{3}{2}, 2\right\}$ or $\left\{0, \frac{1}{2}, 1, \frac{3}{2}\right\}$ as both do not contain the other end point

The n -subintervals into which a partition $p = \{x_0, x_1, x_2, \dots, x_n\}$ divides $[a, b]$ are

$$[x_0, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{k-1}, x_k], \dots, [x_{n-1}, x_n]$$

The length of each sub-interval is $x_1 - x_0, x_2 - x_1, \dots, x_k - x_{k-1}, \dots, x_n - x_{n-1}$

Thus length of the k^{th} - subinterval is $\Delta x_k = x_k - x_{k-1}$; $k = 1, 2, 3, \dots, n$

For the partition $\left\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{2}, 2\right\}$ of $[0,2]$, $\Delta x_1 = x_1 - x_0 = \frac{1}{4}$;

$$\Delta x_2 = x_2 - x_1 = \frac{1}{4}$$

$$\Delta x_3 = x_3 - x_2 = \frac{1}{2} ; \Delta x_4 = x_4 - x_3 = \frac{1}{2}$$

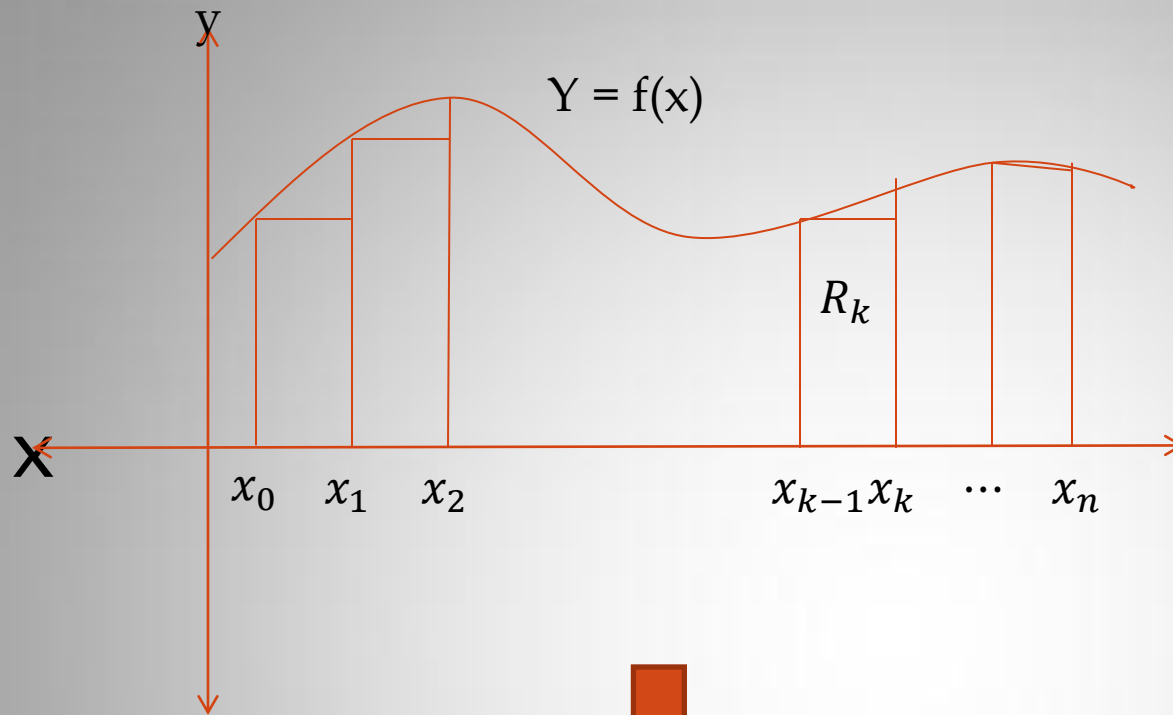
And for the partition $\left\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2\right\}$ of $[0,2]$, $\Delta x_k = \frac{1}{4}$ for each k , $k = 1, 2, \dots, 8$

(Note that $b-a = \Delta x_1 + \Delta x_2 + \dots + \Delta x_n$)

$$\Delta x = \frac{b-a}{n} \text{ (the width of each } n \text{ sub-interval)}$$

Lower sum

Consider the region R bounded by the graph of a non-negative continuous function on $[a,b]$ and the x -axis. Let fill the region with inscribed rectangles



Inscribed rectangles in a region bounded by the graph of $y = f(x)$ in $[a, b]$. Observe that the height of each rectangle is $f(x_{k-1})$ for $k = 1, 2, \dots, n$

Cont...

Consider k^{th} rectangle (R_k). if f is continuous on $[x_{k-1}, x_k]$, then it has max. or min. value on $[x_{k-1}, x_k]$. Let m_k be the smallest value of f in the k^{th} interval so one can see that m_k is the height of the k^{th} rectangle (R_k).

Then the area of the k^{th} rectangle = $m_k \Delta x_k$ and the area of the whole region is

$$\text{Area of } R \approx m_1 \Delta x_1 + m_2 \Delta x_2 + \dots + m_n \Delta x_n$$

Note that the sum of the area of each rectangle does not exceed the area of the region(R)

For the partition $p = \{x_0, x_1, x_2, \dots, x_{k-1}, x_k, \dots, x_n\}$, $m_1 = f(x_0)$, $m_2 = f(x_1)$, $m_3 = f(x_2)$, $\dots$, $m_k = f(x_{k-1})$, $\dots$, $m_n = f(x_{n-1})$

$$L_f(p) = f(x_0)\Delta x_1 + f(x_1)\Delta x_2 + \dots + f(x_{k-1})\Delta x_k + \dots + f(x_{n-1})\Delta x_n$$

$$= \sum_{i=0}^{n-1} f(x_i)\Delta x_{i+1} \quad \text{which is the Lower sum}$$

Example

$$\text{Let } f(x) = x^2, 0 \leq x \leq 2$$

Find $L_f(p)$ and $L_f(p')$ for the following partition

$$p = \left\{0, \frac{1}{2}, 1, \frac{3}{2}, 2\right\} \quad p' = \left\{0, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}, 2\right\}$$

Cont...

Solution

For the partition p

$$m_1 = f(0) = 0; m_2 = f\left(\frac{1}{2}\right) = \frac{1}{4}; m_3 = f(1) = 1; m_4 = f\left(\frac{3}{2}\right) = \frac{9}{4}$$

$$\Delta x_k = \frac{1}{2} \text{ for } k = 1, 2, 3, 4$$

$$L_f(p) = \sum_{i=0}^{n-1} f(x_i) \Delta x_{i+1} = f(0) \frac{1}{2} + f\left(\frac{1}{2}\right) \frac{1}{2} + f(1) \frac{1}{2} + f\left(\frac{3}{2}\right) \frac{1}{2} = \frac{7}{4}$$

and for the partition p'

$$m_1 = f(0) = 0; m_2 = f\left(\frac{1}{4}\right) = \frac{1}{16}; m_3 = f\left(\frac{1}{2}\right) = \frac{1}{4}; m_4 = f(1) = 1; m_5 = f\left(\frac{3}{2}\right) = \frac{9}{4}$$

$$\Delta x_k = \frac{1}{4}, k = 1, 2 \text{ and } \Delta x_k = \frac{1}{2}, k = 3, 4, 5$$

$$L_f(p') = \sum_{i=0}^{n-1} f(x_i) \Delta x_{i+1} = f(0) \frac{1}{4} + f\left(\frac{1}{4}\right) \frac{1}{4} + f\left(\frac{1}{2}\right) \frac{1}{2} + f(1) \frac{1}{2} + f\left(\frac{3}{2}\right) \frac{1}{2} = \frac{113}{64}$$

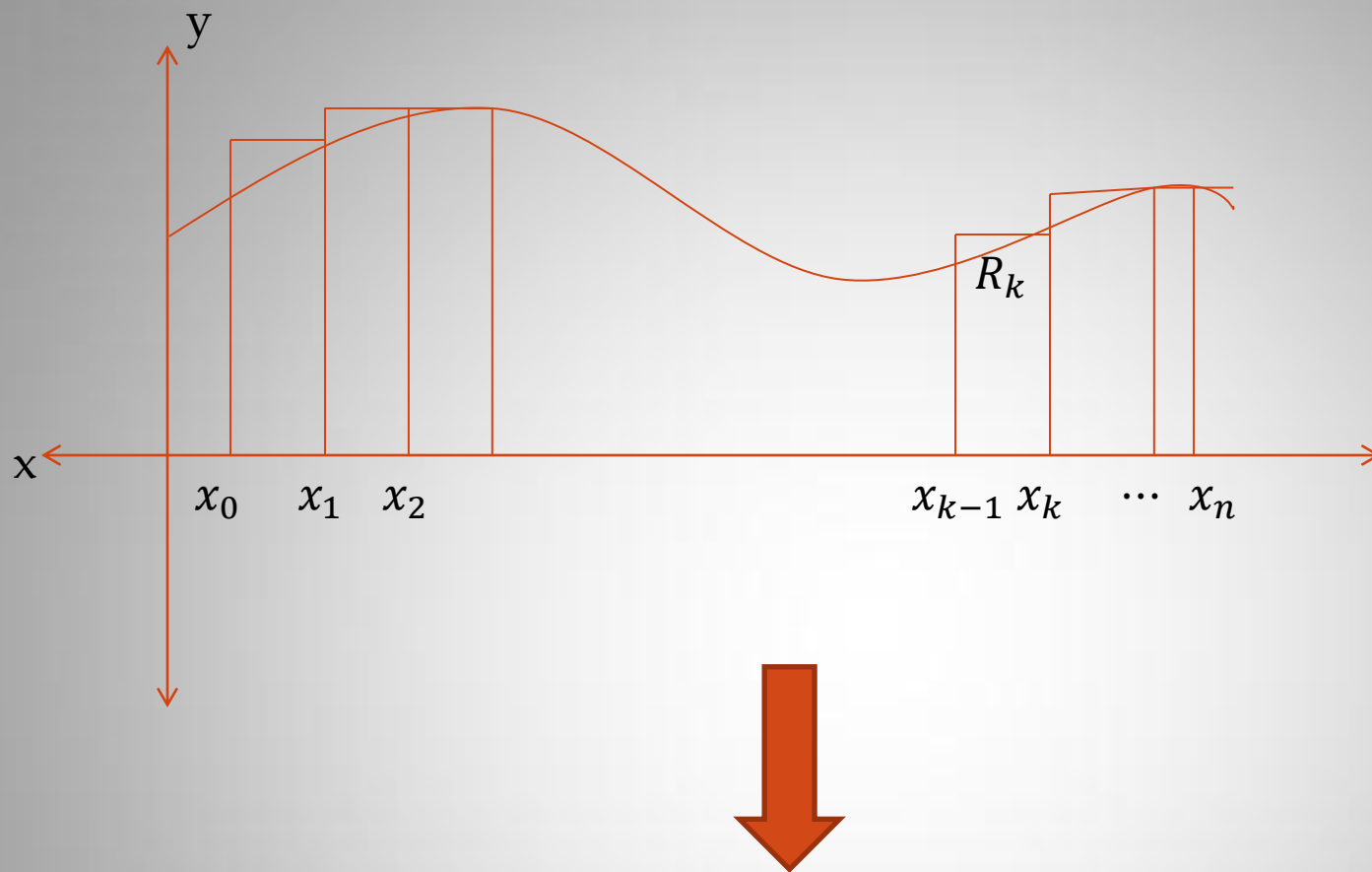
and observe that $L_f(p) < L_f(p')$

Upper Sum

For the case upper Sum, we fill the region with circumscribed rectangle

Cont...

A.H⁰



Inscribed rectangles in a region bounded by the graph of $y = f(x)$ in $[a, b]$. Observe that the height of each rectangle is $f(x_k)$ for $k = 1, 2, 3, \dots, n$

Cont...

A.H

The area of the

k^{th} rectangle $(R_k) = M_k \Delta x_k$, where M_k is the height and Δx_k is the width

Area of $R = M_1 \Delta x_1 + M_2 \Delta x_2 + \dots + M_k \Delta x_k + \dots + M_n \Delta x_n$. For $p = \{x_0, x_1, x_2, \dots, x_k, \dots, x_n\}$, $M_1 = f(x_1), M_2 = f(x_2), \dots, M_k = f(x_k), \dots, M_n = f(x_n)$

$$\begin{aligned}\therefore U_f(p) &= f(x_1)\Delta x_1 + f(x_2)\Delta x_2 + \dots + f(x_k)\Delta x_k + \dots + f(x_n)\Delta x_n \\ &= \sum_{i=1}^n f(x_i)\Delta x_i \quad (\text{upper sum})\end{aligned}$$

Example:

Find $U_f(p)$ and $U_f(p')$. Use the function and the two partition given above

Solution:

For $p = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$; $M_1 = f(\frac{1}{2}) = \frac{1}{4}$; $M_2 = f(1) = 1$; $M_3 = f(\frac{3}{2}) = \frac{9}{4}$; $M_4 = f(2) = 4$

$\Delta x_k = \frac{1}{2}$, $k = 1, 2, 3, 4$, then

$$U_f(p) = \sum_{i=1}^n f(x_i)\Delta x_i = f\left(\frac{1}{2}\right)\frac{1}{2} + f(1)\frac{1}{2} + f\left(\frac{3}{2}\right)\frac{1}{2} + f(2)\frac{1}{2} = \frac{15}{4}$$

and for $p' = \{0, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}, 2\}$; $M_1 = f(\frac{1}{4}) = \frac{1}{16}$; $M_2 = f(\frac{1}{2}) = \frac{1}{4}$; $M_3 = f(1) = 1$

$M_4 = f(\frac{3}{2}) = \frac{9}{4}$; $M_5 = f(2) = 4$; $\Delta x_k = \frac{1}{4}$, $k = 1, 2$ and $\Delta x_k = \frac{1}{2}$, $k = 3, 4, 5$, then

$$U_f(p') = f\left(\frac{1}{4}\right)\frac{1}{4} + f\left(\frac{1}{2}\right)\frac{1}{4} + f(1)\frac{1}{2} + f\left(\frac{3}{2}\right)\frac{1}{2} + f(2)\frac{1}{2} = \frac{237}{64}$$

and Observe that $U_f(p') \leq U_f(p)$

Cont...

If f is any continuous non-negative function on $[a, b]$ and if $p = \{x_0, x_1, \dots, x_n\}$ is an arbitrary partition of $[a, b]$, then m_k and M_k are the min. and max. values of f on $[x_{k-1}, x_k]$ for $k = 1, 2, 3, \dots, n$, so that $m_k \leq M_k$ and this implies $L_f(p) \leq U_f(p)$ and

the area of R is a number between $L_f(p)$ & $U_f(p)$ and one can write

$$L_f(p) < L_f(p') \leq \text{area of } R \leq U_f(p') < U_f(p)$$

As the number of partition (rectangle) increases one can find a better approximation of the area of R

When the n -subdivision of $[a, b]$ gets arbitrary large

$$\text{i.e. } \lim_{n \rightarrow \infty} L_f(p) = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(x_i) \Delta x_{i+1} \text{ and } \lim_{n \rightarrow \infty} U_f(p) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i$$

then each of these limits is the limit of Riemann Sum and denoted by

$\int_a^b f(x) dx$ and this is called definite integral of f from a to b where the integrand

f is continuous function on $[a, b]$ and

$\sum_{i=1}^n f(x_i) \Delta x_i$ is called the Riemann sum

Cont...

Note that the definite integral $\int_a^b f(x)dx$ is a number depending on f , a and b but not x so that the variable appearing in the integral is dummy variable, thus the variable can be replaced by any other variable.

$$\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$$

The definite integral is just a number and it is the area of a region bounded by the graph of f in the given interval $[a, b]$ and the x -axis

$$\text{Area} = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(x_i) \Delta x_{i+1} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i = \int_a^b f(x)dx$$

In general we may have the following definition

Cont...

Definition : If f is a continuous function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subinterval of the endpoints of these subintervals and we choose sample points $x_1^*, x_2^*, x_3^*, \dots, x_n^*$ in those sub intervala so x_i^* lies in i^{th} subinterval $[x_{i-1}, x_i]$. Then the definite integral of f from a to b is

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x \text{ -----}*$$

In the notation $\int_a^b f(x)dx$, $f(x)$ is called the *integrand*, a and b are called the lower and upper limit of the integration respectively, dx indicates the variable of integration.

Note: If we choose the sample points to be the right-end point from each sub-interval, then $x_i^* = x_{i-1}$ and

* will be $\lim_{n \rightarrow \infty} \mathcal{L}_f(p)$ but if $x_i^* = x_i$ then * will be $\lim_{n \rightarrow \infty} U_f(p)$

Cont...

In order to apply the definition given above and evaluate some definite integrals one need to know how to work with sums, so the following three formulas for sum of powers of positive integers help us greatly

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Example:

a. Evaluate the Reimann Sum for $f(x) = x^3 - 6x$ taking the sample points to be right-hand end points and $a=0$ and $n = 6$

b. Evaluate $\int_0^3 (x^3 - 6x) dx$

Solution

a. with $n= 6$ the interval length is $\frac{b-a}{n} = \frac{3-0}{6} = \frac{1}{2}$

and $p = \left\{ \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3 \right\}$ so the Reimann sum is

$$\begin{aligned} \sum_{i=1}^n f(x_i) \Delta x &= f\left(\frac{1}{2}\right) \frac{1}{2} + f(1) \frac{1}{2} + f\left(\frac{3}{2}\right) \frac{1}{2} + f(2) \frac{1}{2} + f\left(\frac{5}{2}\right) \frac{1}{2} + f(3) \frac{1}{2} \\ &= \frac{1}{2} \left(-\frac{23}{8} - 5 - \frac{45}{8} - 4 + \frac{5}{8} + 9 \right) = \frac{1}{2} \left(-\frac{63}{8} \right) = -\frac{63}{16} \end{aligned}$$

b. with n subinterval, we have $\Delta x = \frac{b-a}{n} = \frac{3}{n}$

and $p = \left\{ 0, \frac{3}{n}, \frac{6}{n}, \frac{9}{n}, \dots, \frac{3i}{n}, \dots, 3 \right\}$

Cont...

We use the right-hand point so that $x_i^* = x_i$

$$\Rightarrow \int_0^3 (x^3 - 6x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{3i}{n}\right) \frac{3}{n}$$

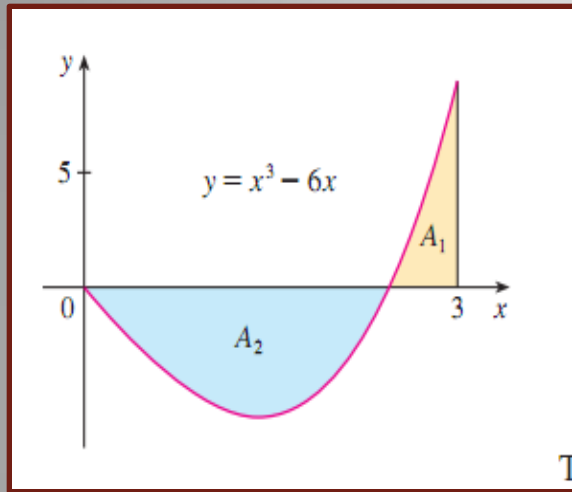
$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right) \right] \frac{3}{n}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \sum_{i=1}^n i^3 - \frac{54}{n^2} \sum_{i=1}^n i \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \left[\frac{n(n+1)}{2} \right]^2 - \frac{54}{n^2} \frac{n(n+1)}{2} \right\}$$

$$= \frac{81}{4} \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 - \frac{54}{2} \lim_{n \rightarrow \infty} \frac{n+1}{n}$$

$$= \frac{81}{4} - 27 = -\frac{27}{4}$$



Note that the value obtained above can be interpreted as the difference of the area of two region $A_1 - A_2$ as shown in the the figure

Cont...

Example: Let $f(x) = c$ for $a \leq x \leq b$, then show that

$$\int_a^b f(x)dx = c(b - a)$$

solution:

Let $p = \{x_0, x_1, x_2, \dots, x_n\}$ be a partition of $[a, b]$ then

$$m_k = c = M_k$$

$$\begin{aligned}\Rightarrow \mathcal{L}_f(p) &= f(x_0)\Delta x_1 + f(x_1)\Delta x_2 + \dots + f(x_{n-1})\Delta x_n \\ &= c(\Delta x_1 + \Delta x_2 + \dots + \Delta x_n) = c(b - a)\end{aligned}$$

$$\begin{aligned}\Rightarrow U_f(p) &= f(x_1)\Delta x_1 + f(x_2)\Delta x_2 + \dots + f(x_n)\Delta x_n \\ &= c(\Delta x_1 + \Delta x_2 + \dots + \Delta x_n) = c(b - a)\end{aligned}$$

$$\therefore \int_a^b f(x)dx = \lim_{n \rightarrow \infty} \mathcal{L}_f(p) \text{ or } \int_a^b f(x)dx = \lim_{n \rightarrow \infty} U_f(p)$$

$$\therefore \int_a^b c dx = c(b - a)$$

Example: Show that

$$\int_a^b x dx = \frac{1}{2}(b^2 - a^2)$$

Cont...

Solution:

Let $P = \{x_0, x_1, x_2, \dots, x_n\}$ be a partition of $[a, b]$. Observe that f is continuous and increasing on $[x_{k-1}, x_k]$, so the min. value of f occurs at x_{k-1} and the max. value occurs at x_k

$$\text{i.e. } m_k = f(x_{k-1}) = x_{k-1}$$

$$M_k = f(x_k) = x_k$$

$$\text{But } m_k = x_{k-1} < \frac{1}{2}(x_{k-1} + x_k) \quad \text{and} \quad M_k = x_k > \frac{1}{2}(x_{k-1} + x_k)$$

$$\mathcal{L}_f(p) = \sum_{i=0}^{n-1} f(x_i) \Delta x_{i+1}, \quad f(x_i) = x_i$$

$$= x_0(x_1 - x_0) + x_1(x_2 - x_1) + \dots + x_{n-1}(x_n - x_{n-1})$$

$$< \frac{1}{2}(x_0 + x_1)(x_1 - x_0) + \frac{1}{2}(x_1 + x_2)(x_2 - x_1) + \dots + \frac{1}{2}(x_{n-1} + x_n)(x_n - x_{n-1})$$

$$= \frac{1}{2}(x_0^2 - x_1^2) + \frac{1}{2}(x_1^2 - x_2^2) + \dots + \frac{1}{2}(x_{n-1}^2 - x_n^2)$$

$$= \frac{1}{2}(x_n^2 - x_0^2) = \frac{1}{2}(b^2 - a^2)$$

Next we find the upper sum $U_f(p)$

$$U_f(p) = \sum_{i=1}^n f(x_i) \Delta x_i \quad ; f(x_i) = x_i$$

Cont...

$$\begin{aligned}
&= x_1(x_1 - x_0) + x_2(x_2 - x_1) + \dots + x_n(x_n - x_{n-1}) \\
&> \frac{1}{2}(x_0 + x_1)(x_1 - x_0) + \frac{1}{2}(x_1 + x_2)(x_2 - x_1) + \dots + \frac{1}{2}(x_{n-1} + x_n)(x_n - x_{n-1}) \\
&= \frac{1}{2}(x_1^2 - x_0^2) + \frac{1}{2}(x_2^2 - x_1^2) + \dots + \frac{1}{2}(x_n^2 - x_{n-1}^2) \\
&= \frac{1}{2}(x_n^2 - x_0^2) = \frac{1}{2}(b^2 - a^2)
\end{aligned}$$

$$= \lim_{n \rightarrow \infty} \mathcal{L}_f(p) \leq \int_a^b x dx \leq \lim_{n \rightarrow \infty} U_f(p)$$

$$\text{But } \lim_{n \rightarrow \infty} \mathcal{L}_f(p) = \frac{1}{2}(b^2 - a^2) = \lim_{n \rightarrow \infty} U_p(p)$$

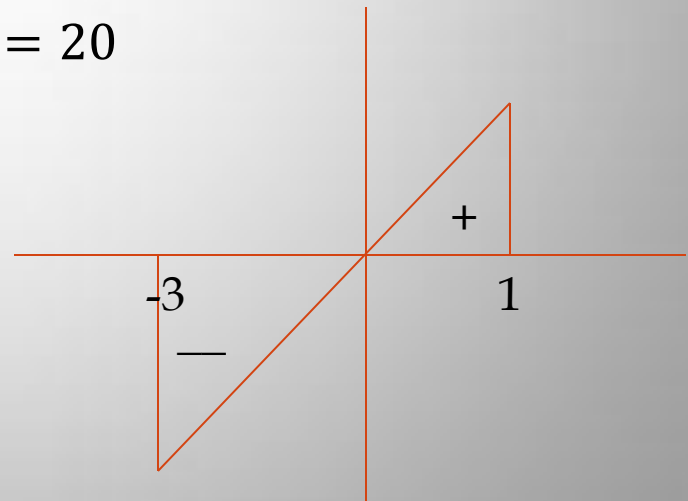
$$\therefore \int_a^b x dx = \frac{1}{2}(b^2 - a^2)$$

Then $\int_{-3}^1 x dx = \frac{1}{2}(1 - 9) = -4$ (the result is the difference of two area)
(see the figure below)

$$\text{If } a = 3 \text{ and } b = 7, \text{ then } \int_3^7 x dx = \frac{1}{2}(49 - 9) = 20$$

Similarly one can show that

$$\int_a^b x^2 dx = \frac{1}{3}(b^3 - a^3)$$



Special Properties of the Definite Integral

- Let f be continuous on $[a, b]$. Then $\int_a^a f(x)dx = 0$ and

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

Example: Evaluate $\int_4^1 x^2 dx$

sol'n

$$\int_4^1 x^2 dx = -\int_1^4 x^2 dx = -\frac{1}{3}(4^3 - 1^3) = -\frac{63}{3} = -21$$

- Let f and g be continuous on $[a, b]$, then

$$\int_a^b [f(x) \pm g(x)]dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

- Let f be continuous and for any numbers a , b , and c

$$\int_a^b cf(x)dx = c \int_a^b f(x)dx$$

- Let f be continuous on an interval containing a , b & c such that $a < c < b$. Then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Cont...

Comparison Properties of definite integral

- a. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x)dx \geq 0$
- b. If $f(x) \leq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$
- c. If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then
$$m(b - a) \leq \int_a^b f(x)dx \leq M(b - a)$$

Theorem (Mean value theorem for Integrals)

Let f be continuous on $[a, b]$, then there is a number c on $[a, b]$ such that

$$\int_a^b f(x)dx = f(c)(b - a)$$

Theorem (Fundamental theorem of calculus for integral)

If f is continuous on the interval $[a, b]$ and $f(t) = F'(t)$, then

$$\int_a^b f(t)dt = F(b) - F(a) \quad (F \text{ is anti derivative of } f)$$

We write $F(b) - F(a) = F(x) \Big|_a^b$

Cont...

Example:

I. Evaluate the following definite integrals

a. $\int_0^2 x^2 dx$ b. $\int_1^4 x^{\frac{1}{2}} dx$ c. $\int_{-1}^1 (1 + 2x) dx$

Solution

a. We use here Evaluating theorem(Fundamental theorem of calculus)

Find first anti derivative of x^2 that is $F(x) = \frac{1}{3}x^3$ then

$$\int_0^2 x^2 dx = F(2) - F(0) = \frac{1}{3}(2)^3 - \frac{1}{3}(0)^3 = \frac{8}{3}$$

b. The anti derivative of $x^{\frac{1}{2}}$ is $F(x) = \frac{2}{3}x^{\frac{3}{2}}$ then

$$\int_1^4 x^{\frac{1}{2}} dx = \frac{2}{3}x^{\frac{3}{2}} \Big|_1^4 = \frac{2}{3}(4)^{\frac{3}{2}} - \frac{2}{3}(1)^{\frac{3}{2}} = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

c. The anti derivative of $1+2x$ is $F(x) = x + x^2$ then $\int_{-1}^1 (1 + 2x) dx$
 $= (x + x^2) \Big|_{-1}^1 = (1 + (-1)^2) - (-1 + (-1)^2) = 2 - 0 = 2$

II. Evaluate the following integrals by interpreting in terms of areas.

a. $\int_{-2}^2 \sqrt{4 - x^2} dx$

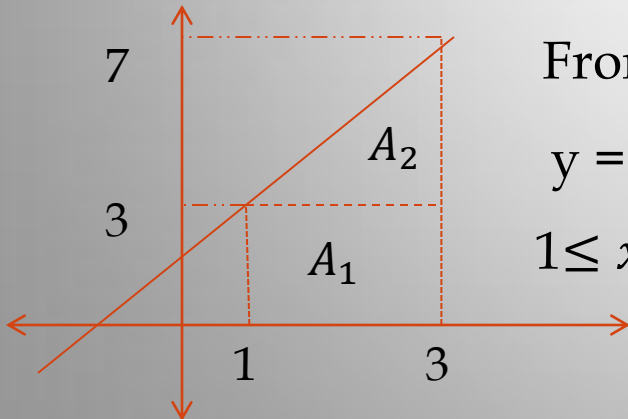
b. $\int_1^3 (1 + 2x) dx$

Solution

a. Here $y = \sqrt{4 - x^2}$; $y \geq 0$ and the integrand represents the semicircle above the x-axis with center (0,0) and radius $r=2$

$$\text{Area} = \int_{-2}^2 \sqrt{4 - x^2} dx = \frac{1}{2} \pi r^2 = 2\pi$$

b. Here $y = 1 + 2x$



From the figure we see that the region below the line $y = 1 + 2x$ and above the x-axis in the interval

$1 \leq x \leq 3$ is divided into two areas A_1 & A_2 such that

$$\text{Area} = \int_1^3 (1 + 2x) dx = A_1 + A_2 = 6 + 4 = 10$$

Exercises

1. Evaluate the Reimann Sum for

a. $f(x) = 2 - x^2$, $0 \leq x \leq 2$ with four subinterval and $x_i^* = x_i$

b. $f(x) = \ln x - 1$, $1 \leq x \leq 4$ with $n = 5$ *and* $x_i^* = x_{i-1}$

2. Use the definition of the definite integral $\int_a^b f(x)dx =$

$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$ to evaluate the following integral

a. $\int_{-1}^5 (1 + 3x)dx$ b. $\int_1^5 (2 + 3x - x^2)dx$ c. $\int_0^5 (1 + 2x^3)dx$

3. Write the sum or difference as a single integral in the form $\int_a^b f(x)dx$

a. $\int_1^3 f(x)dx + \int_3^6 f(x)dx + \int_6^{12} f(x)dx$

b. $\int_2^{10} f(x)dx - \int_2^7 f(x)dx$

4. If $\int_0^{10} f(x)dx = 17$ and $\int_0^8 f(x)dx = 12$ then find $\int_8^{10} f(x)dx$

5. If $\int_0^1 f(t)dt = 2$, $\int_0^4 f(x)dx = -6$ and $\int_3^4 f(x)dx = 1$ then

find $\int_1^3 f(x)dx$

Cont...

Indefinite integrals and their properties

Given a function $f(x)$, an ant derivative of $f(x)$ is any function $F(x)$ such that

$$F'(x) = f(x)$$

If $F(x)$ is any anti derivative of $f(x)$, then the most general anti- derivative of $f(x)$ is called an indefinite integral (or simply integral) and denoted by

$$\int f(x)dx = F(x) + c, \text{ } c \text{ is any constant}$$

Example: Evaluate the following integral

a. $\int x^2 dx$ b. $\int (x^4 + 3x^3 + 2x) dx$

Solution:

a. Let $f(x) = x^2$ We need to find $F(x)$ such that $F'(x) = f(x)$

The general anti-derivative $F(x)$ of $f(x)$ is $\frac{1}{3}x^3 + c$

$$\therefore \int x^2 dx = \frac{1}{3}x^3 + c$$

b. Here also we need to find $F(x)$ such that $F'(x) = f(x)$; $f(x) = x^4 + 3x^3 + 2x$

The general anti-derivative $F(x)$ of $f(x)$ is $\frac{1}{5}x^5 + \frac{3}{4}x^4 + x^2 + c$

Cont...

$$\therefore \int (x^4 + 3x^3 + 2x) dx = \frac{1}{5}x^5 + \frac{3}{4}x^4 + x^2 + c$$

Some common Indefinite Integrals

(here a, c, p, q and r are any arbitrary constants)

$$\int a dx = ax + c$$

$$\int (px + q) dx = \frac{1}{2}px^2 + qx + c$$

$$\int x dx = \frac{1}{2}x^2 + c$$

$$\int x^2 dx = \frac{1}{3}x^3 + c$$

$$\int x^r dx = \frac{1}{r+1}x^{r+1} + c \quad (r \neq -1)$$

$$\int \frac{1}{x} dx = \ln x + c$$

$$\int \sin x dx = -\cos x + c$$

$$\int \cos x dx = \sin x + c$$

$$\int e^x dx = e^x + c$$

Properties of Indefinite Integrals

1. If f and g have anti-derivative on an interval I , then

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

Proof.

Let F and G be the anti-derivative of f and g respectively.

$$\Rightarrow F' = f \quad \text{and} \quad G' = g$$

$$(F + G)' = F' + G' = f + g$$

$\Rightarrow$ The sum of anti derivative of f & g is the ant-derivative of $f + g$
(you can ignore the constant c here)

2. If f has anti-derivative on an interval I and a is a real number, then

$$\int af(x)dx = a \int f(x)dx$$

Proof.

Let F be anti-derivative of f , then $F' = f$

$$(aF)'(x) = aF'(x) = af(x)$$

$\Rightarrow aF$ is the anti-derivative of af

Cont...

Example:

I. Evaluate the indefinite integrals

1. $\int (6x^2 + 5e^x)dx$

2. $\int (5x - 3\cos x)dx$

3. $\int (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0) dx$

4. $\int \sum_{i=1}^n (x^{i+1} + x^i + 2) dx$

Solution:

$$\begin{aligned} 1. \int (6x^2 + 5e^x) dx &= \int 6x^2 dx + \int 5e^x dx \\ &= 6 \int x^2 dx + 5 \int e^x dx \\ &= 2x^3 + 5e^x + c \end{aligned}$$

$$2. \int (5x - 3\cos x) dx = 5 \int x dx - 3 \int \cos x dx = \frac{5}{2} x^2 - 3 \sin x + c$$

$$\begin{aligned} 3. \int [a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0] dx &= a_n \int x^n dx + a_{n-1} \int x^{n-1} dx + \dots \\ &\quad + a_1 \int x dx + a_0 \int dx \\ &= \frac{a_n}{n+1} x^{n+1} + \frac{a_{n-1}}{n} x^n + \dots + \frac{a_1}{2} x^2 + a_0 x + c \end{aligned}$$

Cont...

$$\begin{aligned}
4. \int \sum_{i=1}^n f_i(x) dx &= \sum_{i=1}^n \int (x^{i+1} + x^i + 2) dx \\
&= \sum_{i=1}^n \left[\int x^{i+1} dx + \int x^i dx + \int 2 dx \right] \\
&= \sum_{i=1}^n \left[\frac{1}{i+2} x^{i+2} + \frac{1}{i+1} x^{i+1} + 2x \right] + c \\
&= \sum_{i=1}^n \frac{1}{i+2} x^{i+2} + \sum_{i=1}^n \frac{1}{i+1} x^{i+1} + 2 \times \sum_{i=1}^n 1 + c \\
&= \sum_{i=1}^n \frac{1}{i+2} x^{i+2} + \sum_{i=1}^n \frac{1}{i+1} x^{i+1} + 2xn + c
\end{aligned}$$

II. Evaluate the following indefinite integral.

$$1. \int \sec^2 x dx$$

$$2. \int 2^x dx$$

$$3. \int a^x dx ; (a > 0)$$

Cont...

Solution

1. Let's find the anti derivative of $\sec^2 x$

$$(\tan x)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{1}{\cos^2 x} = \sec^2 x$$

$\Rightarrow \tan x$ is the anti derivative of $\sec^2 x$

$\Rightarrow \tan x + c$ is the general anti – derivative of $\sec^2 x$

$$\Rightarrow \int \sec^2 x dx = \tan x + c$$

2. Let's find the anti derivative of 2^x

$$\text{see that } (2^x)' = 2^x \ln 2$$

$$\Rightarrow \frac{2^x}{\ln 2} \text{ is the anti derivative of } 2^x \text{ as } \left(\frac{2^x}{\ln 2}\right)' = 2^x$$

$$\therefore \int 2^x dx = \frac{2^x}{\ln 2} + c$$

3. Similar to 2 above

$$\int a^x dx = \frac{a^x}{\ln a} + c$$

Cont...

Techniques of Integration

- Integration by Substitution

To evaluate an integral, finding the anti derivative of the integrand function is so important but it may not be always easy to find the anti derivative for an integrand function. So we use some technique to determining the value of an integral. The first technique is integration by substitution which can be explained in the example below

Example: In the integral $\int 6x(3x^2 + 2)^4 dx$

observe the relation b/n $6x$ and $3x^2 + 2$

$$(3x^2 + 2)' = 6x$$

We use change of variable here from variable x to a new variable u

So let $u = 3x^2 + 2$

Cont...

$$\begin{aligned} du = 6x dx & \quad / \quad \therefore \int 6x(3x^2 + 2)^4 dx = \int u^4 du \\ & = \frac{1}{5} u^5 + c \\ & = \frac{1}{5} (3x^2 + 2)^5 + c \end{aligned}$$

Notice that we had to return to the original variable x at the final stage

Let $G'(x) = g(x)$ (G is the anti derivative of g)

$$\frac{d[G(f(x))]}{dx} = G'(f(x))f'(x) = g(f(x))f'(x)$$
$$\Rightarrow \int g(f(x))f'(x)dx = G(f(x)) + c$$

To evaluate such integral we substitute u for inner function $f(x)$ and du for $f'(x)dx$ such that

$$\int g(f(x))f'(x)dx = \int g(u)du = G(u) + c = G(f(x)) + c$$

Cont...

Example:

1. Find $\int 3x^2(x^3 + 5)^6 dx$

Solution

$$\text{Let } u = x^3 + 5$$

$$du = 3x^2 dx$$

$$\Rightarrow \int 3x^2(x^3 + 5)^5 dx = \int u^6 du = \frac{1}{7}u^7 + c = \frac{1}{7}(x^3 + 5)^7 + c$$

2. Find $\int \sin^4 x \cos x dx$

Solution

$$\text{Let } u = \sin x$$

$$du = \cos x dx$$

$$\Rightarrow \int \sin^4 x \cos x = \int u^4 du = \frac{1}{5}u^5 + c = \frac{1}{5}\sin^5 x + c$$

3. Find $\int x^5 \sqrt{x^2 - 1} dx$

Solution

$$\text{Let } u = \sqrt{x^2 - 1}$$

Cont...

A.H

$$du = \frac{xdx}{\sqrt{x^2-1}} = \frac{xdx}{u} \Rightarrow udu = xdx \Rightarrow \text{the integrand function is written as}$$

$$x^4 x \sqrt{x^2-1} dx = x^4 u^2 du, \text{ so we must express } x^4 \text{ in terms of } u$$

$$u = \sqrt{x^2-1} \Rightarrow u^2 + 1 = x^2 \Rightarrow u^4 + 2u^2 + 1 = x^4$$

$$\begin{aligned} \Rightarrow \int x^5 \sqrt{x^2-1} dx &= \int x^4 u^2 du = \int (u^4 + 2u^2 + 1) u^2 du \\ &= \int (u^6 + 2u^4 + u^2) du = \int u^6 du + 2 \int u^4 du + \int u^2 du \\ &= \frac{1}{7} u^7 + \frac{2}{5} u^5 + \frac{1}{3} u^3 + c \\ &= \frac{1}{7} (x^2 - 1)^{\frac{7}{2}} + \frac{2}{5} (x^2 - 1)^{\frac{5}{2}} + \frac{1}{3} (x^2 - 1)^{\frac{3}{2}} + c \end{aligned}$$

4. Evaluate the integral given below by substitution

$$\int (x-1)\sqrt{x+1} dx$$

Solution

$$\text{Let } u = x+1 ; du = dx$$

$$\begin{aligned} \int (x-1)\sqrt{x+1} dx &= \int (u-2)\sqrt{u} du \quad (\text{just by adding } -2 \text{ to } u = x+1) \\ &= \int (u\sqrt{u} - 2\sqrt{u}) du \end{aligned}$$

Cont.

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} + c$$

$$= \frac{2}{5} (x+1)^{\frac{5}{2}} - \frac{4}{3} (x+1)^{\frac{3}{2}} + c$$

5. Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$ (this is definite integral)

Solution

Let $u = 3-5x$, then $du = -5dx \Rightarrow dx = -\frac{du}{5}$

When $x = 1$, $u = -2$ and when $x=2$, $u = -7$

Thus $\int_1^2 \frac{dx}{(3-5x)^2} = -\frac{1}{5} \int_{-2}^{-7} \frac{du}{u^2} = -\frac{1}{5} \left(-\frac{1}{u} \Big|_{-2}^{-7} = \frac{1}{5u} \Big|_{-2}^{-7} = \frac{1}{5} \left(-\frac{1}{7} + \frac{1}{2} \right) = \frac{1}{14} \right.$

6. Find the area of the region bounded by $y = (2 + \sin \frac{x}{2})^2 \cos \frac{x}{2}$, the x axis, and the lines $x = 0$ and $x = \pi$

Solution

Here observe that $y \geq 0$, when $0 \leq x \leq \pi$, the required area is

$$\int_0^\pi (2 + \sin \frac{x}{2})^2 \cos \frac{x}{2} dx$$

Let $u = 2 + \sin \frac{x}{2}$, $du = \frac{1}{2} \cos \frac{x}{2} dx$

Cont.

$$\text{Area} = 2 \int_{u(0)}^{u(\pi)} u^2 du = \frac{2}{3} u^3 \Big|_2^3 = \frac{2}{3} (27 - 8) = \frac{38}{3} \text{ square units}$$

Exercises

I. Evaluate the integral by making the given substitution

1. $\int x(4 + x^2)^{10} dx$, $u = 4 + x^2$
2. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$, $u = \sqrt{x}$
3. $\int e^{\sin \theta} \cos \theta d\theta$, $u = \sin \theta$
4. $\int x^2 2^{x^3+1} dx$, $u = x^3 + 1$

II. Evaluate the indefinite integral by appropriate substitution

1. $\int \frac{(\ln x)^2}{x} dx$
2. $\int \frac{x+1}{\sqrt{x^2+2x+3}} dx$
3. $\int \sqrt{\cot x} \csc^2 x dx$
4. $\int \tan^2 x \sec^2 x dx$
5. $\int \frac{\tan^{-1} x}{1+x^2} dx$
6. $\int \frac{3x-1}{(3x^2-2x+1)^4} dx$
7. $\int \frac{x}{(x-5)^6} dx$
8. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$
9. $\int (2x+3)\sqrt{2x-1} dx$
10. Find the function whose tangent has slope $x\sqrt{x^2+5}$ for each value of x and whose graph passes through the point $(2,5)$

III. Evaluate the definite integral by appropriate substitution

1. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$
2. $\int_0^4 x^3 (x^2 + 1)^{-\frac{1}{2}} dx$
3. $\int_{\frac{\pi^2}{16}}^{\frac{\pi^2}{9}} \frac{2 \sin \sqrt{x} \cos \sqrt{x}}{\sqrt{x}} dx$
4. $\int_0^{\frac{\pi}{2}} \cos^3 x dx$

Integration By Parts

Let u and v be variables depending on x and both be differentiable. Then

$$\frac{d(uv)}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} \quad \text{by integrating both side we'll have}$$

$$uv = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx = \int v du + \int u dv$$

$$\Rightarrow \int u dv = uv - \int v du \quad ; \text{if } u = f(x) \wedge v = g(x), \text{ then we can}$$

also write integration by parts as follows

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Example: I. Find $\int x \cos x dx$

Solution:

$$\text{Let } u = x ; du = dx$$

$$dv = \cos x dx ; v = \sin x$$

$$\begin{aligned} \int u dv &= \int x \cos x dx = uv - \int v du = x \sin x - \int \sin x dx \\ &= x \sin x + \cos x + c \end{aligned}$$

Cont...

How to choose u and dv in using integration by parts

1. Whatever we take for dv, we need to be able to find v
 2. Whatever we take for u , du needs to be simpler than u
 3. whatever we take for dv, v needs to be simpler than dv
- for example

if $dv = x$, then $v = \frac{1}{2}x^2$ which is more complicated than dv

and if $dv = \ln x$, then $v = x \ln x - x$ which is more complicated than dv so we must take care when we select for u and dv

2. Find $\int \ln x dx$

solution

$$\text{Let } u = \ln x ; du = \frac{1}{x} dx$$

$$dv = dx ; v = x$$

$$\int \ln x dx = uv - \int u dv = x \ln x - \int \frac{1}{x} (x) dx = x \ln x - x + c$$

3. Find $\int x^2 \sin 4x dx$

Solution

Cont...

$$\text{Let } u = x^2 ; \quad du = 2x$$

$$dv = \sin 4x ; \quad v = -\frac{1}{4} \cos 4x$$

$$\int u dv = \int x^2 \sin 4x = uv - \int v du = x^2 \left(-\frac{1}{4} \cos 4x \right) + \frac{1}{2} \int x \cos 4x dx$$

here we need to applying the same technique on $\int x \cos 4x dx$ one more time so,

$$\text{Let } u = x ; \quad du = dx$$

$$dv = \cos 4x ; \quad v = \frac{1}{4} \sin 4x$$

$$\int u dv = \int x \cos 4x dx = uv - \int v du = \frac{1}{4} x \sin 4x - \frac{1}{4} \int \sin 4x dx$$

$$= \frac{1}{4} x \sin 4x + \frac{1}{16} \cos 4x$$

$$\therefore \int x^2 \sin 4x dx = -\frac{1}{4} x^2 \cos 4x + \frac{1}{8} x \sin 4x + \frac{1}{32} \cos 4x + c$$

3. Find $\int e^{2x} \sin 3x dx$ (use integration by parts)

Solution

$$\text{Let } u = e^{2x} ; \quad du = 2e^{2x} dx$$

$$dv = \sin 3x dx ; \quad v = -\frac{1}{3} \cos 3x$$

Cont...

$$\int u dv = \int e^{2x} \sin 3x dx = uv - \int v du = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} \int e^{2x} \cos 3x dx$$

Here we need to apply by parts once again on $\int e^{2x} \cos 3x dx$

so let $u = e^{2x}$; $du = 2e^{2x} dx$

$$dv = \cos 3x \quad ; \quad v = \frac{1}{3} \sin 3x$$

$$\int u dv = \int e^{2x} \cos 3x dx = uv - \int v du = \frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \int e^{2x} \sin 3x dx$$

$$\therefore \int e^{2x} \sin 3x dx = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{9} e^{2x} \sin 3x - \frac{4}{9} \int e^{2x} \sin 3x dx$$

$$\Rightarrow \frac{13}{9} \int e^{2x} \sin 3x dx = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{9} e^{2x} \sin 3x$$

$$\Rightarrow \int e^{2x} \sin 3x dx = -\frac{3}{13} e^{2x} \cos 3x + \frac{2}{13} e^{2x} \sin 3x + c$$

Exercises

I. Evaluate the integral (use integration by parts)

$$1. \int \frac{\ln x}{x^3} dx \quad 2. \int x^3(x^2 + 1) dx \quad 3. \int \frac{x}{\sqrt{x+2}} dx \quad 4. \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x \csc^2 x dx$$

$$5. \int_0^{\frac{\pi}{2}} (\ln x)^3 dx \quad 6. \int_1^4 x (\ln x)^3 dx$$

Cont...

A.H

Integration by Partial fraction

In this section we are concerned with integral of the form

$\int \frac{P(x)}{Q(x)} dx$, where $P(x)$ and $Q(x)$ are polynomial and $\frac{P(x)}{Q(x)}$ is called rational function. We see here only when the degree of $P(x)$ is less than the degree of $Q(x)$

The first thing to do in the evaluation of the integral $\int \frac{P(x)}{Q(x)} dx$ is

expressing $\frac{P(x)}{Q(x)}$ as sums of simple fractions, called partial fraction(decomposition)

Let's use the following example to explain all about this section

Example: I. Express the following fractions in partial fraction and then evaluate their indefinite integral.

$$1. \frac{3x+7}{x^2+6x+8}$$

$$2. \frac{x+1}{(x-1)^2(x-2)}$$

$$3. \frac{x^2-1}{(2x+3)(x^2+1)}$$

Solution

Cont...

$$1. \frac{3x+7}{x^2+6x+8} = \frac{3x+7}{(x+2)(x+4)} = \frac{A}{x+2} + \frac{B}{x+4} \Rightarrow A(x+4) + B(x+2) = 3x+7$$

If $x = -2$ then B vanishes and then $2A = 1 \Rightarrow A = \frac{1}{2}$

If $x = -4$ then A vanishes and then $-2B = -5 \Rightarrow B = \frac{5}{2}$

$$\therefore \frac{3x+7}{x^2+6x+8} = \frac{\frac{1}{2}}{x+2} + \frac{\frac{5}{2}}{x+4}$$

$$\Rightarrow \int \frac{3x+7}{x^2+6x+8} dx = \frac{1}{2} \int \frac{1}{x+2} dx + \frac{5}{2} \int \frac{1}{x+4} dx ; \text{ Let } u = x+2 \quad ; du = dx$$

and $v = x+4 \quad ; dv = dx$

We do integration by substitution.

$$\begin{aligned} \int \frac{3x+7}{x^2+6x+8} dx &= \frac{1}{2} \int \frac{1}{u} du + \frac{5}{2} \int \frac{1}{v} dv = \frac{1}{2} \ln|u| + \frac{5}{2} \ln|v| + c \\ &= \frac{1}{2} \ln|x+2| + \frac{5}{2} \ln|x+4| + c \end{aligned}$$

$$2. \frac{x+1}{(x-1)^2(x-2)} = \frac{A}{(x-1)^2} + \frac{B}{x-1} + \frac{C}{x-2}$$

$$\Rightarrow A(x-2) + B(x-1)(x-2) + C(x-1)^2 = x+1$$

Now let $x = 1$ then B and C vanishes and then $A = -2$

let $x = 2$ then A and B vanishes and then $C = 3$

let $x = 0$ then with some calculation , $B = -3$

$$\frac{x+1}{(x-1)^2(x-2)} = \frac{-2}{(x-1)^2} + \frac{-3}{x-1} + \frac{3}{x-2}, \text{ then}$$

$$\begin{aligned} \int \frac{x+1}{(x-1)^2(x-2)} dx &= -2 \int \frac{1}{(x-1)^2} dx - 3 \int \frac{1}{x-1} + 3 \int \frac{1}{x-2} dx \\ &= \frac{2}{x-1} - 3 \ln|x-1| + 3 \ln|x-2| \end{aligned}$$

$$3. \frac{x^2-1}{(2x+3)(x^2+1)} = \frac{A}{2x+3} + \frac{Bx+C}{x^2+1} \Rightarrow A(x^2+1) + (Bx+C)(2x+3) = x^2-1$$

Cont...

From the equation above we may have

$$A + 2B = 1$$

$$3B + 2C = 0 \Rightarrow$$

$$3C + A = 1$$

From the first two equation we get
 $-3A + 4C = -3$ and with the third equation we

$$\text{get } A = \frac{5}{13} ; B = \frac{4}{13} ; C = -\frac{6}{13}$$

$$\Rightarrow \frac{x^2-1}{(2x+3)(x^2+1)} = \frac{\frac{5}{13}}{2x+3} + \frac{\frac{4}{13}x - \frac{6}{13}}{x^2+1} = \frac{\frac{5}{13}}{2x+3} + \frac{\frac{4}{13}x}{x^2+1} - \frac{\frac{6}{13}}{x^2+1}$$

$$\begin{aligned} \therefore \int \frac{x^2-1}{(2x+3)(x^2+1)} dx &= \frac{5}{13} \int \frac{1}{2x+3} dx + \frac{4}{13} \int \frac{x}{x^2+1} dx - \frac{6}{13} \int \frac{1}{x^2+1} dx \\ &= \frac{5}{26} \ln |2x+3| + \frac{2}{13} \ln |x^2+1| - \frac{6}{13} \tan^{-1} x + c \end{aligned}$$

Note that when the degree of $P(x)$ is greater than the degree of $Q(x)$ we use long division first and then do the integral using one of the technique we know (see the example given below)

Example: Let $r(x) = \frac{x^3}{x^2-3x+2}$, then to find its integral we first do

long division

Cont...

A.H

$\frac{x^3}{x^2-3x+2} = x + 3 + \frac{7x-6}{x^2-3x+2}$; But we must find the partial fraction of $\frac{7x-6}{x^2-3x+2} = \frac{-1}{x-1} - \frac{8}{x-2}$

$$\Rightarrow \int \frac{x^3}{x^2-3x+2} dx = \frac{1}{2}x^2 + 3x - \ln|x-1| - 8\ln|x-2|$$

Exercises

I. Evaluate the integral given below

$$1. \int \frac{x-9}{x^2+3x-10} dx \quad 2. \int \frac{x^2+2x-1}{x^3-x} dx \quad 3. \int \frac{10}{(x-1)(x^2+9)} dx$$

$$4. \int \frac{1}{x^3-x} dx \quad 5. \int \frac{x^2+2}{4x^5+4x^3+x} dx \quad 6. \int \frac{dx}{e^{2x}-4e^x+4} dx$$

II. Evaluate the definite integral (use long division first)

$$1. \int_0^1 \frac{x^3}{x^2+1} dx \quad 2. \int_0^2 \frac{x^3+x^2-12x+1}{x^2+x+2} dx$$

Cont...

Trigonometric Substitution

In integration of algebraic function that contains an expression of the form $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$, the best way to do such integration is to make a trigonometric substitution that gets rid of the root sign.

For $\sqrt{a^2 - x^2}$, ($a > 0$), we substitute $x = a \sin \theta$

$$\Rightarrow \sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = a \sqrt{1 - \sin^2 \theta} = a \cos \theta$$

For $\sqrt{a^2 + x^2}$ or $\frac{1}{x^2 + a^2}$ ($a > 0$), we substitute $x = a \tan \theta$

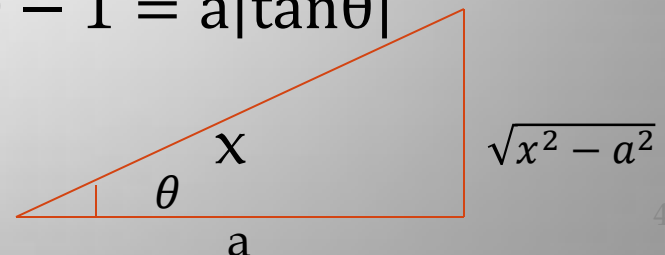
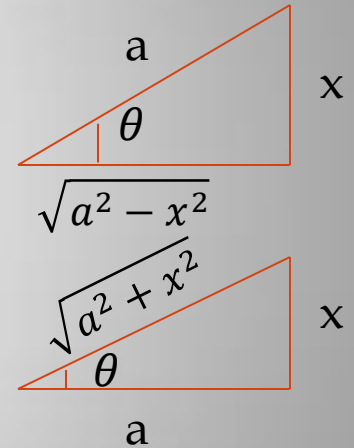
$$\Rightarrow \sqrt{a^2 + x^2} = \sqrt{a^2 + a^2 \tan^2 \theta} = a \sqrt{1 + \tan^2 \theta} = a \sec \theta$$

For $\sqrt{x^2 - a^2}$, ($a > 0$), we substitute $x = a \sec \theta$

$$\Rightarrow \sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = a \sqrt{\sec^2 \theta - 1} = a |\tan \theta|$$

$$(\sqrt{x^2 - a^2} = \begin{cases} a \tan \theta ; x \geq a \\ -a \tan \theta ; x \leq -a \end{cases})$$

Cont...



Example: Evaluate the following integrals(use trigonometric Substitution)

$$1. \int \frac{dx}{x\sqrt{1-4x^2}}$$

$$2. \int \frac{dx}{(5-x^2)}$$

$$3. \int \frac{(x+1)dx}{\sqrt{9-x^2}}$$

Solution

$$1. \text{ Let } x = \frac{1}{2}\sin\theta$$

$$dx = \frac{1}{2}\cos\theta d\theta$$

$$\begin{aligned} \Rightarrow \int \frac{dx}{x\sqrt{1-4x^2}} &= \int \frac{(\cos\theta/2)d\theta}{(\sin\theta/2)\sqrt{1-\sin^2\theta}} \\ &= \int \csc\theta d\theta \\ &= \ln|\csc\theta - \cot\theta| + c \\ &= \ln\left|2x - \frac{\sqrt{1-4x^2}}{2x}\right| + c \end{aligned}$$

$$2. \text{ Let } x = \sqrt{5}\sin\theta$$

$$dx = \sqrt{5} \cos\theta d\theta$$

$$\begin{aligned} \Rightarrow \int \frac{dx}{(5-x^2)} &= \int \frac{\sqrt{5} \cos\theta d\theta}{(5-5\sin^2\theta)} = \frac{\sqrt{5}}{5} \int \sec\theta d\theta \\ &= \frac{\sqrt{5}}{5} \ln|\sec\theta + \tan\theta| + c \\ &= \frac{\sqrt{5}}{5} \ln\left|\frac{\sqrt{5}+x}{\sqrt{5-x^2}}\right| + c \end{aligned}$$

Cont...

A.H

$$\begin{aligned}
3. \text{ Let } x &= 3\sin\theta & \Rightarrow \int \frac{(x+1)dx}{\sqrt{9-x^2}} &= \int \frac{(3\sin\theta+1)}{\sqrt{9-9\sin^2\theta}} 3\cos\theta d\theta \\
dx &= 3\cos\theta & &= \int (3\sin\theta + 1)d\theta \\
& & &= -3\cos\theta + \theta + C \\
& & &= -\sqrt{9-x^2} + \sin^{-1}\left(\frac{x}{3}\right) + c
\end{aligned}$$

II. Evaluate the following integrals(use trigonometric Substitution)

$$1. \int \frac{dx}{(1+2x^2)^{5/2}}$$

$$2. \int \frac{\sqrt{9+x^2}}{x^4} dx$$

$$3. \int \frac{dx}{x^2+x+1}$$

Solution

$$\begin{aligned}
1. \text{ Let } x &= \frac{1}{\sqrt{2}}\tan\theta & \Rightarrow \int \frac{dx}{(1+2x^2)^{5/2}} &= \int \frac{1/\sqrt{2} \sec^2\theta d\theta}{(1+\tan^2\theta)^{5/2}} = \frac{1}{\sqrt{2}} \int \cos^3\theta d\theta \\
dx &= \frac{1}{\sqrt{2}} \sec^2\theta & &= \frac{1}{\sqrt{2}} \left(\sin\theta - \frac{1}{3}\sin^3\theta \right) + c \\
& & &= \frac{x}{\sqrt{1+2x^2}} - \frac{2x^3}{3(1+2x^2)^{3/2}} + c
\end{aligned}$$

Cont...

2. Let $x = 3\tan\theta$

$$dx = 3\sec^2\theta d\theta$$

$$\begin{aligned}\Rightarrow \int \frac{\sqrt{9+x^2}}{x^4} dx &= \int \frac{\sqrt{9+9\tan^2\theta}}{81\tan^4\theta} 3\sec^2\theta d\theta \\ &= \frac{1}{9} \int \frac{\sec^4\theta}{\tan^4\theta} d\theta = \frac{1}{9} \int \csc^4\theta d\theta \\ &= -\frac{1}{27} \cot\theta \csc^2\theta - \frac{2}{27} \cot\theta \csc\theta + c \\ &= -\frac{3}{27x} \left(\frac{9+x^2}{9}\right) - \frac{2}{9x^2} \sqrt{9+x^2} + c\end{aligned}$$

3. We apply here perfect square technique so we write the denominator as follows

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\Rightarrow \int \frac{dx}{x^2+x+1} = \int \frac{dx}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}} \quad \text{so, Let } x+\frac{1}{2} = \frac{\sqrt{3}}{2}\tan\theta \quad ; dx = \frac{\sqrt{3}}{2}\sec^2\theta d\theta$$

$$\begin{aligned}\int \frac{dx}{x^2+x+1} &= \int \frac{dx}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}} = \int \frac{\frac{\sqrt{3}}{2}\sec^2\theta d\theta}{\frac{3}{4}\tan^2\theta+\frac{3}{4}} = \frac{2\sqrt{3}}{3} \int d\theta = \frac{2\sqrt{3}}{3} \theta \\ &= \frac{2\sqrt{3}}{3} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + c\end{aligned}$$

III. Evaluate the following integrals (use trigonometric Substitution)

1. $\int \frac{dx}{3x^2-9}$

2. $\int \frac{dx}{\sqrt{9x^2-4}}$

3. $\int \frac{dx}{x^2\sqrt{x^2-a^2}}$

Solution

1. Let $x = \sqrt{3} \sec\theta$

$$dx = \sqrt{3} \tan\theta \sec\theta d\theta$$

$$\begin{aligned}\Rightarrow \int \frac{dx}{3x^2-9} &= \int \frac{\sqrt{3} \tan\theta \sec\theta d\theta}{9 \sec^2\theta - 9} = \frac{\sqrt{3}}{9} \int \frac{\tan\theta \sec\theta}{\tan^2\theta} d\theta \\ &= \frac{\sqrt{3}}{9} \int \frac{d\theta}{\sin\theta} = \frac{\sqrt{3}}{9} \int \csc\theta d\theta \\ &= \frac{\sqrt{3}}{9} \ln|\csc\theta - \cot\theta| + c\end{aligned}$$

$$= \frac{\sqrt{3}}{9} \ln \left| \sqrt{\frac{x-\sqrt{3}}{x+\sqrt{3}}} \right| + c = -\frac{1}{6\sqrt{3}} \ln \left| \frac{x+\sqrt{3}}{x-\sqrt{3}} \right| + c$$

2. Let $x = \frac{2}{3} \sec\theta$

$$dx = \frac{2}{3} \tan\theta \sec\theta d\theta$$

$$\begin{aligned}\Rightarrow \int \frac{dx}{\sqrt{9x^2-4}} &= \int \frac{2/3 \tan\theta \sec\theta d\theta}{\sqrt{4 \sec^2\theta - 4}} = \frac{1}{3} \int \frac{d\theta}{\sin\theta} = \frac{1}{3} \int \csc\theta d\theta \\ &= \frac{1}{3} \ln|\csc\theta - \cot\theta| + c\end{aligned}$$

$$= \frac{1}{3} \ln \left| x + \sqrt{x^2 - \frac{4}{9}} \right| + c$$

3. Let $x = a \sec\theta$

$$dx = a \tan\theta \sec\theta d\theta$$

$$\begin{aligned}\Rightarrow \int \frac{dx}{x^2 \sqrt{x^2 - a^2}} &= \int \frac{a \tan\theta \sec\theta d\theta}{a^2 \sec^2\theta \sqrt{a^2 \sec^2\theta - a^2}} \\ &= \frac{1}{a^2} \int \frac{d\theta}{\sec\theta} = \frac{1}{a^2} \int \cos\theta d\theta = \frac{1}{a^2} \sin\theta + c \\ &= \frac{1}{a^2} \frac{\sqrt{x^2 - a^2}}{x} + c\end{aligned}$$

Cont...

A.H

Exercises

Evaluate the following integral using appropriate trigonometric substitution.

$$1. \int \frac{dx}{\sqrt{x^2-25}}$$

$$2. \int \frac{\sqrt{1-x^2}}{x^3} dx$$

$$3. \int \sqrt{x^2+1} dx$$

$$4. \int \frac{\sqrt{9x^2-1}}{x^2} dx$$

$$5. \int \frac{dx}{x^2\sqrt{x^2+4}}$$

$$6. \int \frac{dx}{x^2+6x+14}$$

$$7. \int_0^2 \frac{(x^2+12)dx}{x^2+4}$$

$$8. \int_2^3 \frac{dx}{x^2-1}$$

$$9. \int_1^2 \frac{dx}{(x^2-2x+4)^{3/2}}$$

Cont.

Improper Integral

Definition of an improper integral of type I

a. If $\int_a^t f(x)dx$ exists for every number $t \geq a$, then

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

provided that the limit exists

b. If $\int_t^b f(x)dx$ exists for every number $t \leq b$, then

$$\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx$$

provided that the limit exists

Notice that the improper integral $\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exist.

c. If both $\int_a^\infty f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we

$$\text{define } \int_{-\infty}^\infty f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^\infty f(x)dx$$

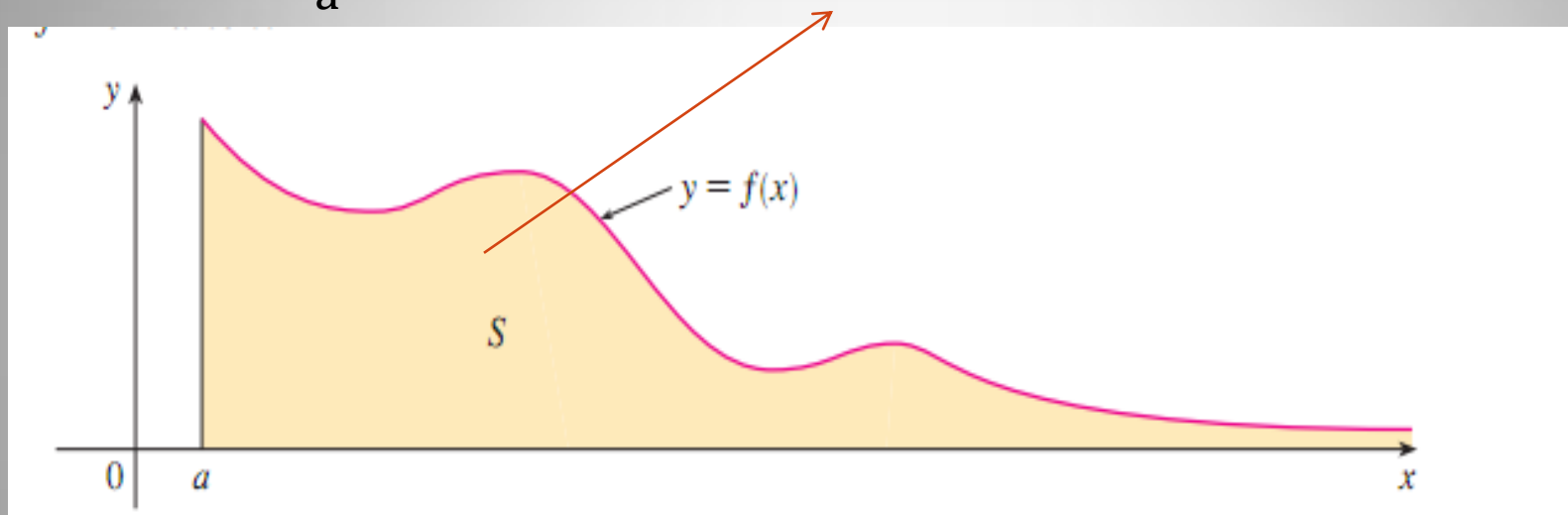
Cont...

A.H

Any of the improper integral above can be interpreted as an area provided that f is a positive function.

If $\int_a^\infty f(x)dx$ is convergent, then the area of the region S is

$$A(S) = \int_a^\infty f(x)dx \quad , \quad \text{where } S = \{(x,y): x \geq a, 0 \leq y \leq f(x)\}$$



Example:

1. Determine whether the integral $\int_1^\infty f(x)dx$ is convergent or divergent

Solution

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} \ln|x| \Big|_1^t = \lim_{t \rightarrow \infty} (\ln t - \ln 1) = \lim_{t \rightarrow \infty} \ln t = \infty$$

$\Rightarrow$ The limit does not exist

$\therefore \int_1^\infty \frac{1}{x} dx$ is divergent (check that $\int_1^\infty \frac{1}{x^2} dx$ is convergent (finite number))

Cont...

2. Evaluate $\int_{-\infty}^0 x e^x dx$

Solution

$$\int_{-\infty}^0 x e^x dx = \lim_{t \rightarrow -\infty} \int_t^0 x e^x dx$$

$$\text{Let } u = x, du = dx$$

$$dv = e^x dx, v = e^x$$

$$\int_t^0 x e^x dx = x e^x \Big|_t^0 - e^x \Big|_t^0 = -t e^t - 1 + e^t$$

$$\begin{aligned} \int_{-\infty}^0 x e^x dx &= \lim_{t \rightarrow -\infty} \left(-\frac{t}{x^{-t}} \right) + \lim_{t \rightarrow -\infty} (-1) + \lim_{t \rightarrow -\infty} e^t \\ &= 0 + 0 - 1 = -1 \quad (\text{we use here l'Hospital's rule}) \end{aligned}$$

3. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

solution

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2}$$

Let's evaluate each integral on the left side separately

Cont...

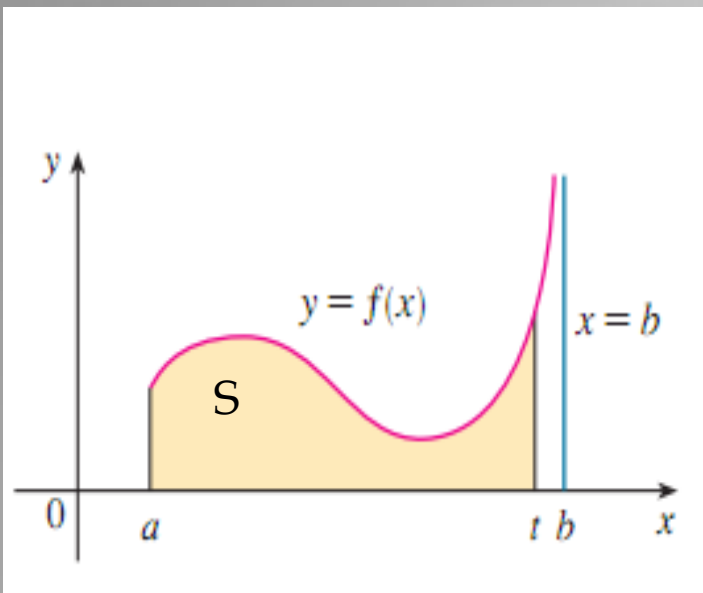
A.H

$$\begin{aligned}
\int_0^{\infty} \frac{dx}{1+x^2} &= \lim_{t \rightarrow \infty} \int_0^t \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \tan^{-1} x \Big|_0^t \\
&= \lim_{t \rightarrow \infty} (\tan^{-1} t - \tan^{-1} 0) = \lim_{t \rightarrow \infty} \tan^{-1} t = \frac{\pi}{2} \\
\int_{-\infty}^0 \frac{dx}{1+x^2} &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \tan^{-1} x \Big|_t^0 \\
&= \lim_{t \rightarrow -\infty} (\tan^{-1} 0 - \tan^{-1} t) = - \lim_{t \rightarrow -\infty} \tan^{-1} t = \frac{\pi}{2} \\
\therefore \int_{-\infty}^{\infty} \frac{dx}{1+x^2} &= 2\left(\frac{\pi}{2}\right) = \pi
\end{aligned}$$

Type 2: Discontinuous Integrands

Suppose that f is a positive continuous function defined on a finite interval $[a, b)$, but has a vertical asymptote at b so that the region is infinite in a vertical direction as shown in the figure below.

Cont...



The area of the part of S between a and t is $A(S) = \int_a^t f(x)dx$

$A(S)$ approaches to a definite number as $t \rightarrow b^-$, then

$$A(S) = \int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

Definition of an Improper Integral of Type 2

a. If f is continuous on $[a, b]$ and is discontinuous at b , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

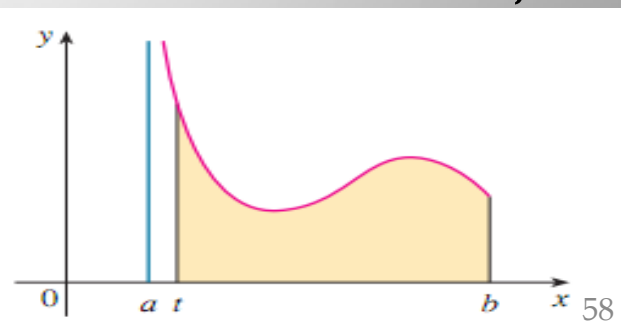
provided that the limit exists

b. If f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$

provided that the limit exists

Cont...



The improper integral $\int_a^b f(x)dx$ is called *convergent* if the corresponding limit exists and *divergent* if the limit does not exist

c. If f has a discontinuity at c , where $a < c < b$, and both

$\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then we define

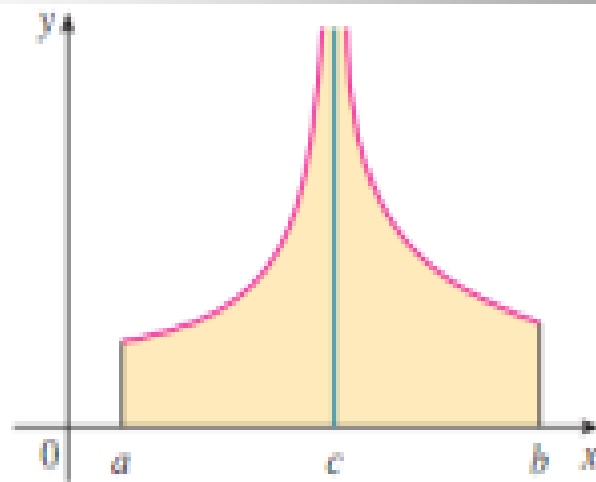
$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad (\text{see the figure below})$$

Example

1. Evaluate $\int_2^5 \frac{dx}{\sqrt{x-2}}$

Solution

since $f(x) = \frac{1}{\sqrt{x-2}}$ has vertical asymptote at $x = 2$, then the



given integral is improper(f is continuous on $(2, 5]$, but not at $x = 2$, so we use definition b

$$\begin{aligned}\Rightarrow \int_2^5 \frac{dx}{\sqrt{x-2}} &= \lim_{t \rightarrow 2^+} \int_t^5 \frac{dx}{\sqrt{x-2}} = \lim_{t \rightarrow 2^+} 2\sqrt{x-2} \Big|_t^5 \\ &= \lim_{t \rightarrow 2^+} 2(\sqrt{3} - \sqrt{t-2}) = 2\sqrt{3}\end{aligned}$$

Exercises

1. For what values of p is the integral

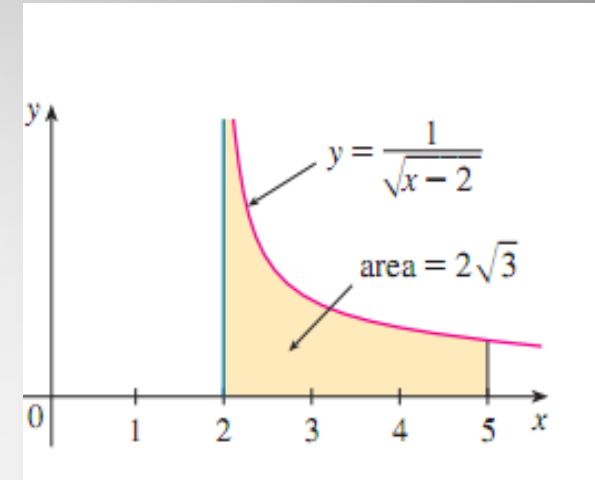
$$\int_1^{\infty} \frac{dx}{x^p} \text{ convergent}$$

2. Determine whether $\int_0^{\frac{\pi}{2}} \sec x dx$ convergent?

3. Evaluate $\int_0^3 \frac{dx}{x-1}$

(be careful here, write the integral as a sum of two integral in $0 \leq 1 \leq 3$)

4. Determine whether each integral is convergent or divergent



Cont...

A.H

a. $\int_1^{\infty} \frac{dx}{(3x+1)^2}$

b. $\int_{-\infty}^{-1} e^{-2t}$

c. $\int_{-\infty}^{\infty} x^3 dx$

d. $\int_{-\infty}^{\infty} \frac{dx}{x^2+4}$

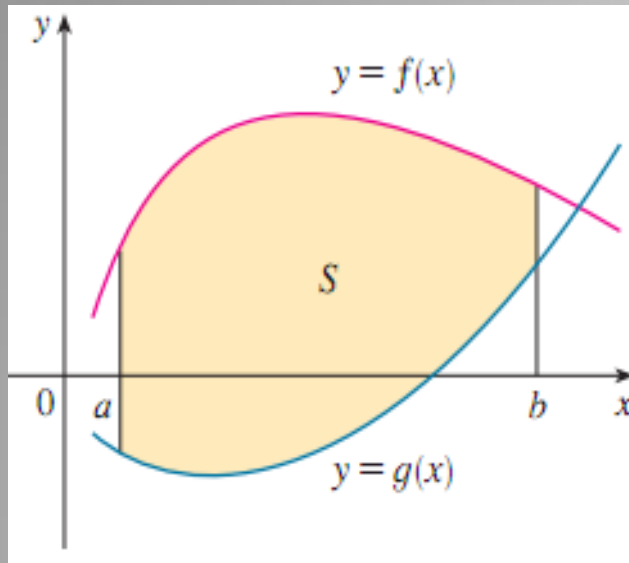
e. $\int_0^1 \frac{dy}{4y-1}$

Application of Integrals

Areas between curves

Consider the region S that lies between two curves $y = f(x)$, $y = g(x)$

and between the vertical lines $x = a$ and $x = b$ where f & g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$ (see the figure below)



$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x$$

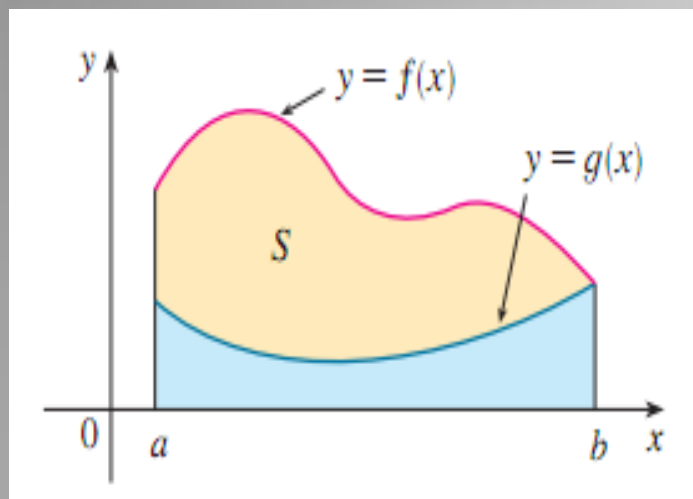
we know that the limit is the definite integral of $f-g$, then we have the following formula for the area of region b/n two curves

The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$ and the lines $x=a$, $x = b$, where f & g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$ is

$$A = \int_a^b [f(x) - g(x)] dx$$

Cont...

A.H



If $g(x) = 0$, S is the region under the graph of f and our definition for area reduced to what we knew before i.e $A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$

In the case where both f & g are positive, you can see from the figure above that

$$\begin{aligned} A &= [\text{Area under } f(x)] - [\text{Area under } g(x)] \\ &= \int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b [f(x) - g(x)] dx \end{aligned}$$

Example

1. Find the area of the region enclosed by the parabolas

Cont...

$$f(x) = 2x - x^2 \quad \text{and} \quad g(x) = x^2$$

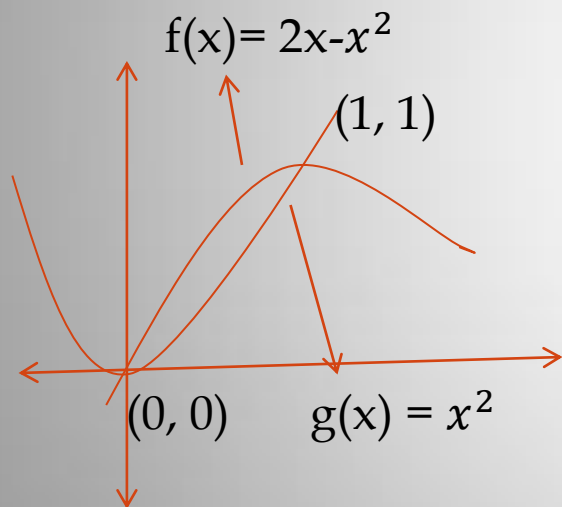
Solution

We must first find points of intersection of the two curves

$$2x - x^2 = x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow 2x(x - 1) = 0$$

$$\Rightarrow x = 0 \text{ and } x = 1$$

Therefore the points of intersection are $(0, 0)$ & $(1, 1)$



From the figure we observe that $f(x) \geq g(x)$

$$\Rightarrow A = \int_0^1 [f(x) - g(x)] dx$$

$$= \int_0^1 (2x - 2x^2) dx = 2 \int_0^1 (x - x^2) dx$$

$$= 2 \left(\frac{1}{2} x^2 - \frac{1}{3} x^3 \right) \Big|_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$

Some region S are best treated by regarding x as a function of y . If a region is bounded by curves with equation $x = f(y)$, $x = g(y)$, $y = c$ and $y = d$, where f and g are continuous and

$f(y) \geq g(x)$ for $c \leq y \leq d$, then the area is

$$A = \int_c^d [f(y) - g(y)] dy$$

2. Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$

Solution

We must first find the points of intersection of the two curves

$$(x-1)^2 = 2x + 6 \Rightarrow x^2 - 2x + 1 = 2x + 6$$

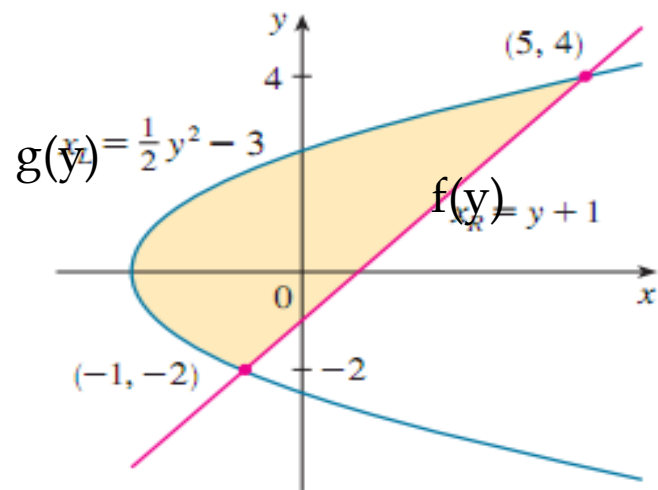
$$\Rightarrow x^2 - 4x - 5 = 0 \Rightarrow (x + 1)(x - 5) = 0$$

$$\Rightarrow x = -1, x = 5$$

Therefore the points of intersection are $(-1, -2)$ & $(5, 4)$

Lets write the function of the two curve as $f(y) = y + 1$ and

$$g(y) = \frac{1}{2}y^2 + 3$$



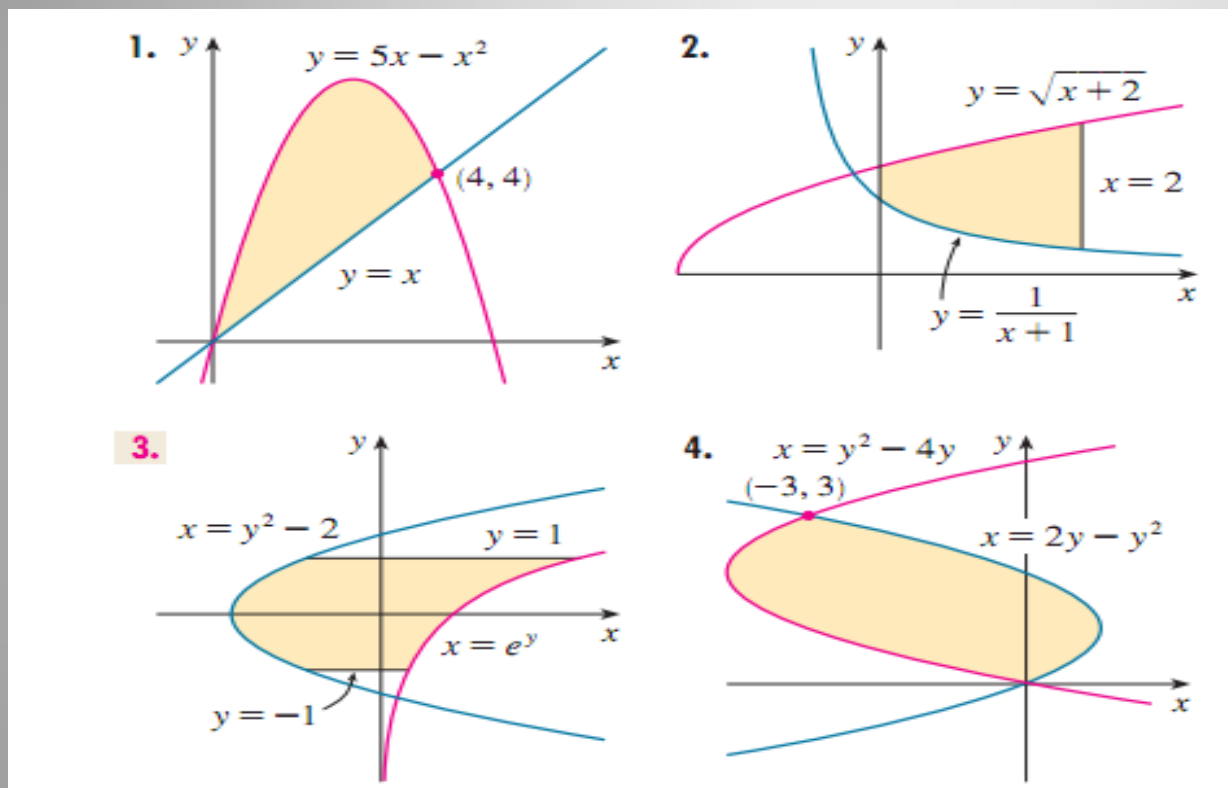
From the figure we observe that $f(y)$ is to the right of $g(y)$ so that $f(y) \geq g(y)$ for $-2 \leq y \leq 4$. Then the area of the region b/n the two curves is

$$\begin{aligned}
 A &= \int_{-2}^4 [f(y) - g(y)] dy = \int_{-2}^4 ((y + 1) - (\frac{1}{2}y^2 - 3)) dy \\
 &= \int_{-2}^4 \left(-\frac{1}{2}y^2 + y + 4\right) dy \\
 &= \left(-\frac{1}{6}y^3 + \frac{1}{2}y^2 + 4y\right) \Big|_{-2}^4 \\
 &= -\frac{35}{6} + \frac{12}{2} + 24 = \frac{145}{6}
 \end{aligned}$$

Cont...

Exercises.

- I. Find the area of the region enclosed by the two curves given in the following graphs



Cont...

A.H

II. Find the area enclosed by the graphs of the two functions given below

a. $y^2 = x$, $x - 2y = 3$

b. $y^2 = 1 - x$, $y^2 = x + 1$

c. $y = |x|$, $y = x^2 - 2$

In case you need the reference books used in preparing this pp, here they are

- James Stewart, Calculus concepts and contexts 2nd ed.
- Robert A. Adams, calculus: Complete course 5th ed.
- Robert. C. Wrede, Murray Spiegel, Advanced calculus(schaum's outlines) 2nd ed.

Thisisthe End of Ch - 6 So does the course